DDCA • FINAL PROJECT

4-BIT MAGNITUDE COMPARATOR

A > B | A = B | A < B

Design and simulation of a combinational logic circuit

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SECTION: 4

LOGIC DESIGN
SIMULATION • ANALYSIS

Introduction & Objective

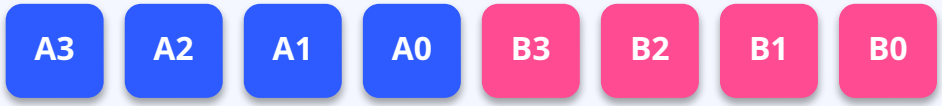
WHAT IS A MAGNITUDE COMPARATOR?

A combinational circuit that compares two binary numbers and indicates whether A is greater than, equal to, or less than B.

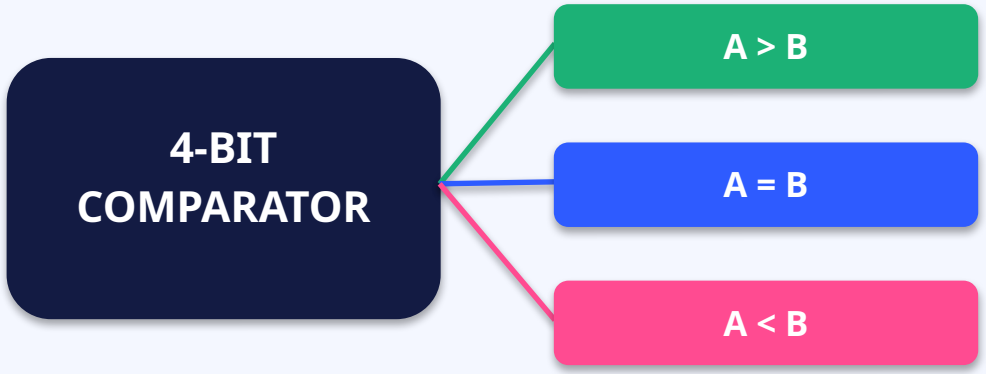
PROJECT OBJECTIVE

Design and test a 4-bit comparator with inputs A3–A0 and B3–B0 and three outputs: A>B, A=B, and A<B.

EIGHT INPUT BITS



Inputs: two 4-bit unsigned binary numbers.
Outputs: exactly one relation is HIGH.



Three mutually exclusive outputs

Theory & Logic Equations

COMPARISON RULE

Start at the most significant bit (MSB). The first unequal bit determines which number is larger.

EQUALITY RULE

XNOR outputs 1 when two bits match. A=B only when all four bit pairs match.

DEFINE BIT MATCHES

$$X_i = A_i \text{ XNOR } B_i$$

$$A = B \rightarrow E = X_3 \cdot X_2 \cdot X_1 \cdot X_0$$

$$A > B \rightarrow G = A_3\bar{B}_3 + X_3A_2\bar{B}_2 + X_3X_2A_1\bar{B}_1 + X_3X_2X_1A_0\bar{B}_0$$

$$A < B \rightarrow L = \bar{A}_3B_3 + X_3\bar{A}_2B_2 + X_3X_2\bar{A}_1B_1 + X_3X_2X_1\bar{A}_0B_0$$

Legend: · = AND | + = OR | bar = NOT | XNOR = equality detector

Circuit Design in Logisim

BUILD STEPS

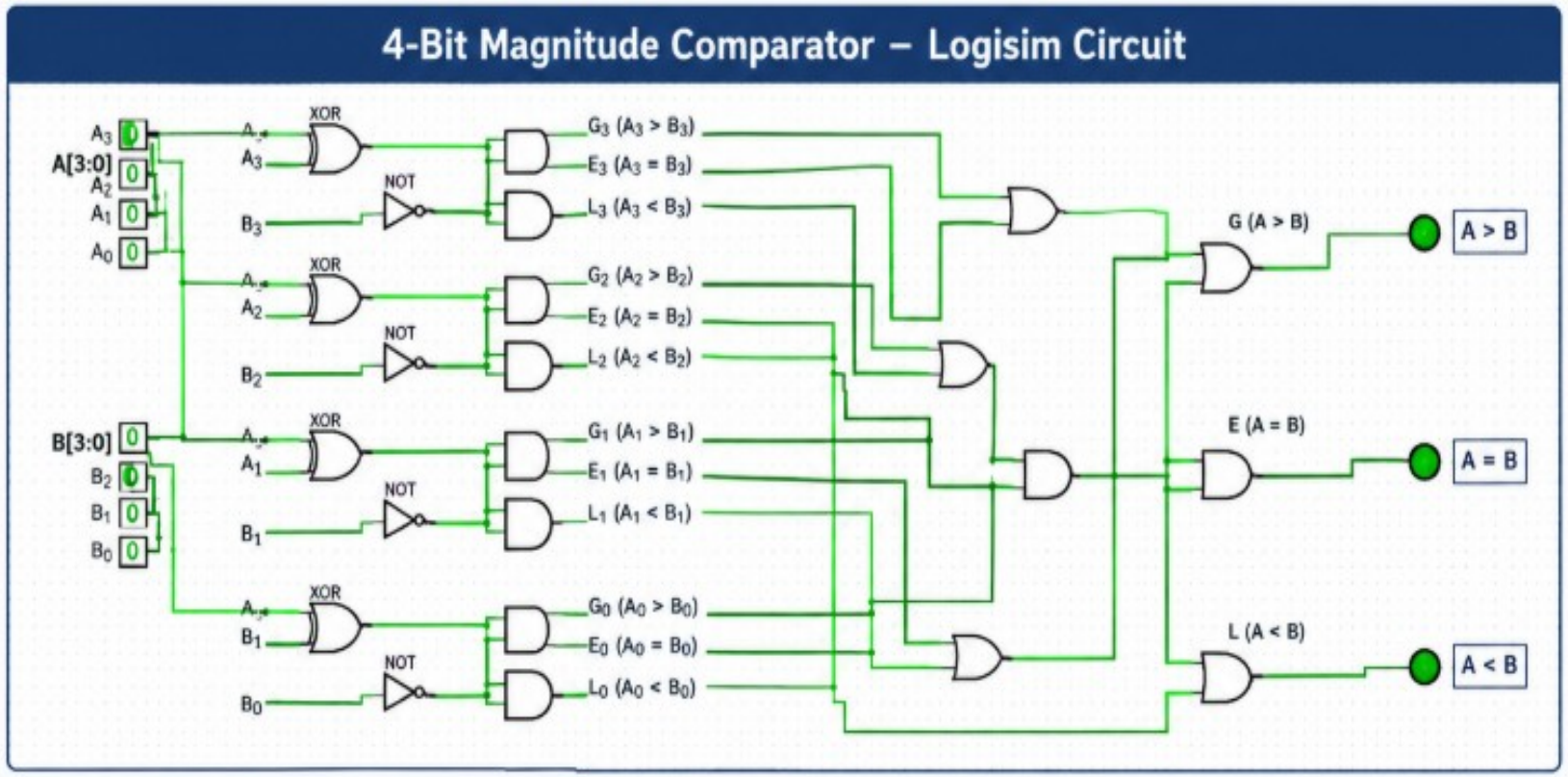
1. Place 8 input pins: A3–A0 and B3–B0.

2. Add four XNOR gates to produce X3–X0.

3. AND all X bits for A=B.

4. Use NOT, AND and OR gates to implement A>B and A<B equations.

5. Connect three output pins.



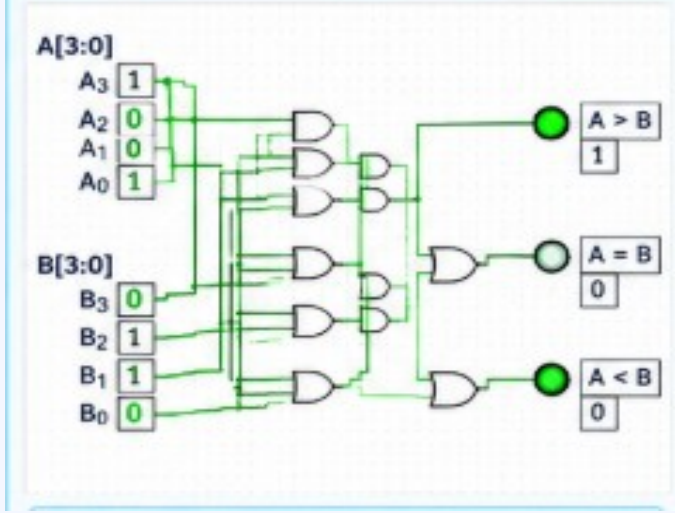
- Compares two 4-bit binary inputs.
- Uses XNOR, AND, and OR gates.
- Produces A > B, A = B, or A < B.

Simulation Results & Truth Table

A	B	A>B	A=B	A<B
1001 (9)	0110 (6)	1	0	0
0101 (5)	0101 (5)	0	1	0
0011 (3)	1010 (10)	0	0	1
1111 (15)	0000 (0)	1	0	0
0000 (0)	0000 (0)	0	1	0

Expected result: exactly one output is HIGH for each input pair.

Logisim Simulation (Example)



Key Observation: The circuit accurately compares two 4-bit inputs and produces the correct output: A > B, A = B, or A < B.

Conclusion & Applications

PROCESSORS

Comparison in ALUs and decision logic.

ADDRESS CHECKING

Compare addresses and numeric ranges.

CONTROL SYSTEMS

Threshold checks and digital control.

CONCLUSION

The 4-bit magnitude comparator compares two unsigned binary numbers and activates the correct output. The truth table and Logisim tests verify its operation

FUTURE IMPROVEMENTS

Extend to 8 or 16 bits, reduce gate delay, or create a reusable hierarchical design.

THANK YOU